Tuesday Warm-up: Electrostatics

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September 11, 2025

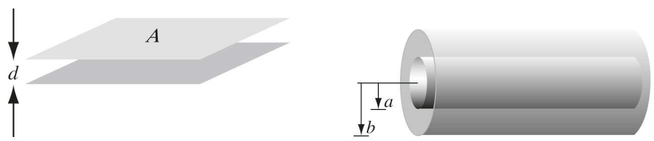

Figure 1: (Left) A parallel-plate capacitor. (Right) A cylindrical coaxial cable with capacitance per unit length.

1 Memory Bank

- Gauss' Law, integral form

$$\oint \mathbf{E} \cdot d\mathbf{a} = \frac{1}{\epsilon_0} Q_{\text{enc}} \quad (1)$$

- Gauss' Law, differential form

$$\nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \rho \quad (2)$$

- Definition of potential

$$V(\mathbf{r}) = -\int_{\mathcal{O}}^{\mathbf{r}} \mathbf{E}(\mathbf{r}') \cdot d\mathbf{l}' \quad (3)$$

- Relationship between potential and field:

$$\mathbf{E} = -\nabla V \quad (4)$$

- Electrostatic energy, point charges:

$$W = \frac{1}{2} \sum_{i=1}^{n} q_i V(\mathbf{r}_i) \quad (5)$$

- Electrostatic energy, continuous charge distribution:

$$W = \frac{1}{2} \epsilon_0 \int E^2 d\tau \quad (6)$$

- Capacitance, C:

$$C = Q/V \quad (7)$$

2 Gauss' Law and Scalar Potential

1. Suppose the voltage (scalar potential) of a charge distribution obeys

$$V(\mathbf{r}) = A \frac{e^{-\lambda r}}{r} \quad (8)$$

where A and λ are constants. Find $\mathbf{E}(\mathbf{r})$, $\rho(r)$, and the total charge, Q.

3 Electrostatic Energy

1. What is the electrostatic energy of an electric dipole, with a charge $+q$ at $(d/2)\hat{\mathbf{x}}$, and a charge $-q$ at $-(d/2)\hat{\mathbf{x}}$?

4 Capacitance

1. Consider the conductors in Fig. 1 (Left). Show that the capacitance is $C = \epsilon_0 A/d$.

2. Consider the conductors in Fig. 1 (Right). Show that the capacitance per unit length is $C/l = 2\pi\epsilon_0/\ln(b/a)$.